


# **L13a**: Synthetic Markets with Jump Hidden Markov Models

CHEME 5660 Cornell University

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# Objectives for Today

Today, we build synthetic markets: simulated price histories with the features of real ones. How would a trading method do on thousands of plausible paths?

## Today's concepts

- Markov chains and hidden Markov models: states, transition matrices, stationary distributions, emissions.
- A jump hidden Markov model of one asset: binned states, counted transitions, Student-t emissions, jump episodes.
- A synthetic market of many stocks: shared market and sector draws plus a draw of each stock's own residual.

# What You Need to Know

## Core concepts

- A Markov chain is a set of states $\mathcal{S}$ and a transition matrix $\mathbf{P}$ whose rows sum to one; its stationary distribution satisfies $\bar{\pi} = \bar{\pi}\mathbf{P}$.
- The jump hidden Markov model bins the daily growth rate into states, estimates $\mathbf{P}$ by counting, and draws each day from a Student-t of the day's state. Jump episodes (probability $\epsilon$, mean length $\lambda$) keep volatility clustering.
- A synthetic stock is $g_i = \alpha_i + \beta_i\, g_m + \delta_i\, f_s + \psi_i\, \tilde{\varepsilon}_i$: shared market and sector draws make stocks move together.

## For understanding (you won't reproduce these)

- The fitting and sampling algorithms of the jump hidden Markov model.
- The tuning objective for $\epsilon$ and $\lambda$, and the shape of an episode.
- How the sector loadings and own-draw scale are chosen, and a drifting $\beta$.

# Lectures and Examples

**Lecture:**
CHEME-5660-L13a-Lecture-JumpHMM-SyntheticMarkets-Fall-2026.ipynb

**Examples:**
CHEME-5660-L13a-Example-JumpHMM-SPY-Fall-2026.ipynb
CHEME-5660-L13a-Example-JumpHMM-SectorFactor-Fall-2026.ipynb

Optional advanced notebooks are listed on the Optional Advanced Material frame.

# Review: Wolfram Markets

Previously, each trader in a Wolfram market followed a rule: a table from the sum of its four neighbors' intentions to its next intention. A small change in that table can turn a calm market into a crash.

[Placeholder: panels of simulated Wolfram markets]

**Today:** our goal is a trade management system that makes money when it can and, at a minimum, preserves cash when it cannot. The system has to adapt as the market changes, and we need many plausible paths of a changing market to test it.

# Company Profile: Two Sigma

Two Sigma: a quantitative investment manager founded in 2001 by David Siegel and John Overdeck, based in New York. In 2021, its researchers let the data define market regimes:

- Four regimes: a Gaussian mixture model fit to 17 drivers of returns, 1971–2020: Steady State (47%), Inflation (20%), Crisis (17%), and Walking on Ice (16%).
- Quick switches: in 2020, Crisis in February and March, then Walking on Ice, then Steady State.
- No memory: the model sorts each period on its own, and the authors say “this model is not predictive.”

**Connection:** the regimes play the role of our hidden states. Our transition matrix adds the probability of moving from today’s mood to tomorrow’s.

# Why Synthetic Markets?

A backtest runs a method on one historical path, and a method that did well may have been lucky. A synthetic market generates many plausible paths. For stress tests, its paths need:

- Fat tails: large daily moves are far more common than a normal distribution predicts.
- Volatility clustering: large moves follow large moves, so the size of the daily move stays correlated for weeks.
- Co-movement: stocks move together, through the market and their sectors.

Geometric Brownian motion can have co-movement through correlated shocks, but neither fat tails nor volatility clustering.

# Markov Chains

A Markov chain is a stochastic (random) process that moves between states. The next state depends only on the present state. Example: the moods of an investor, bullish, neutral, or bearish.

**Definition:** a Markov chain has states $\mathcal{S} = \{s_1, \dots, s_N\}$ and a transition matrix $\mathbf{P} \in \mathbb{R}^{N \times N}$. With $X_t$ the state on day $t$, entry $p_{ij}$ is the probability of moving from $s_i$ today to $s_j$ tomorrow:

$$p_{ij} = P\left(X_{t+1} = s_j \mid X_t = s_i\right)$$

Each row of $\mathbf{P}$ sums to one and is a categorical distribution over tomorrow's state. The stationary distribution $\bar{\pi}$ solves $\bar{\pi} = \bar{\pi}\mathbf{P}$ and gives the long-run share of time the chain spends in each state.

# Estimating the Transition Matrix

We estimate $\mathbf{P}$ from data. **Proposition:** in a sequence of observed states $X_1, \ldots, X_T$, let $n_{ij}$ count the days in state $s_i$ followed by a day in state $s_j$. The maximum likelihood estimate of $p_{ij}$ is the share of the days in state $s_i$ that moved to state $s_j$:

$$\boxed{\hat{p}_{ij} = \frac{n_{ij}}{\sum_{k=1}^{N} n_{ik}}}$$

This estimate maximizes the log-likelihood $\sum_{i,j} n_{ij} \ln p_{ij}$, subject to each row of $\mathbf{P}$ summing to one.

# Hidden Markov Models

A hidden Markov model adds observable outputs to the chain. **Definition:** the states $X_t$ of the chain are hidden. Each day, we observe an output (emission) drawn from a distribution that depends on that day's hidden state.

For a market generator, the hidden state sets how volatile the market is today, and the emission turns the state into a growth rate. A generator draws the first state of each path from $\bar{\pi}$.

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**Full version:** the Markov models notebook (Markov property, power iteration, sampling)

# The Jump Hidden Markov Model

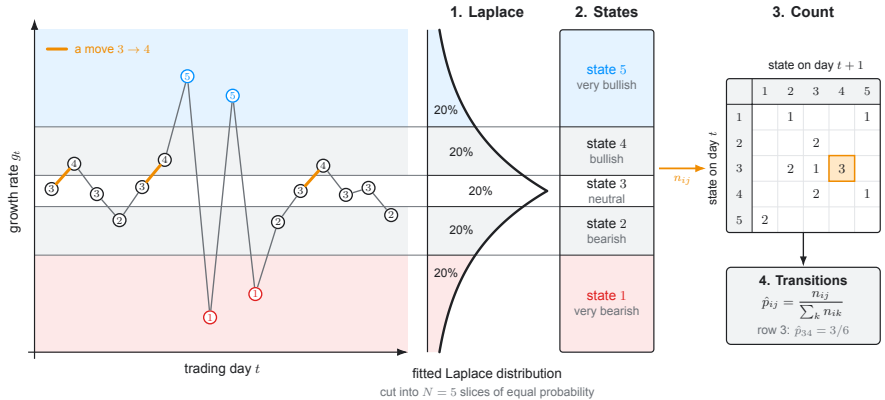
For the daily growth rate $g_t = \frac{1}{\Delta t}\ln(S_t/S_{t-1})$ (1/yr) of one asset:

- States: fit a Laplace distribution to the growth rates and cut it into $N$ bins of equal probability; state 1 holds the most negative growth rates.
- Transitions: assign each day to its bin and count, $\hat{p}_{ij}$.
- Emissions: in state $s_k$, $g_t \mid X_t = s_k \sim \mu_k + \sigma_k T_\nu$: $\mu_k$, $\sigma_k$ the mean and standard deviation of the days in that state, $T_\nu$ a Student-t with $\nu$ fixed.

In the data the states are observed bins; in simulation they are hidden, and only the growth rates are reported.

# Fitting the Model

Each state is a slice of the fitted Laplace distribution and a mood of the market. We count the moves between states:



# Algorithm: Fit the Model

Given the daily growth rates $g_1, \ldots, g_T$ of one ticker, the number of states $N$, and $\nu$:

1. Laplace: $\mu \leftarrow$ the median of the growth rates, and $b \leftarrow \frac{1}{T}\sum_t |g_t - \mu|$.
2. States: $X_t \leftarrow s_k$, where $(k-1)/N \le F_L(g_t) < k/N$.
3. Count: $n_{ij} \leftarrow n_{ij} + 1$ for each move from $X_{t-1} = s_i$ to $X_t = s_j$.
4. Transitions: $\hat{p}_{ij} \leftarrow n_{ij} / \sum_k n_{ik}$.
5. Emissions: $\mu_k$, $\sigma_k \leftarrow$ the mean and standard deviation of the growth rates of the days in state $s_k$.
6. Start: compute the stationary distribution $\bar{\pi}$ of $\hat{\mathbf{P}}$.

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**Full version:** the lecture notebook

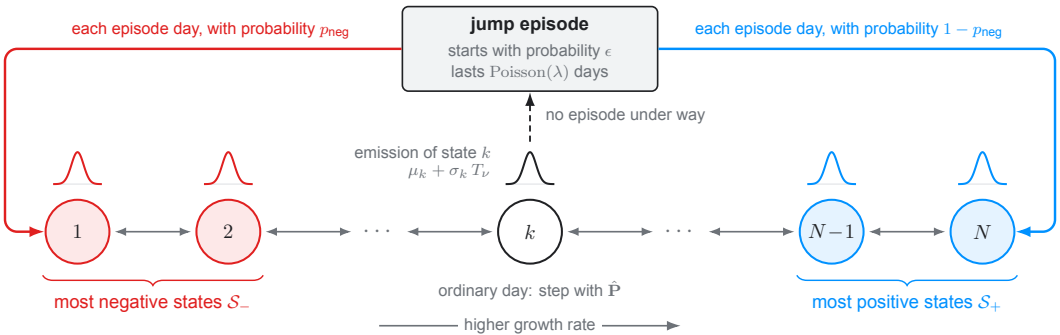
# Why Add Jumps?

The counted chain forgets quickly: after a volatile day, a few transitions usually return it to the middle states, so $|g_t|$ is correlated for only a few days. Jump episodes keep volatile periods going:

- With no episode under way, an episode starts with probability $\epsilon$.
- It lasts a Poisson number of days with mean $\lambda$.
- Each episode day is drawn from the few most extreme states, negative or positive: a run of large moves of both signs.

# The Jump Hidden Markov Model: One Day

An ordinary day steps with $\hat{\mathbf{P}}$. An episode day draws its state from the most negative or the most positive states:



# Algorithm: Sample a Path

Draw $X_1 \sim \bar{\pi}$ and $g_1$ from its emission, and set $\texttt{remaining} \leftarrow 0$. Each day $t = 2, \ldots, T$ runs in this order:

1. New episode: if $\texttt{remaining} = 0$, draw $u \sim \text{Uniform}(0, 1)$. If $u < \epsilon$, draw $\texttt{remaining} \sim \text{Poisson}(\lambda)$.
2. Episode day: if $\texttt{remaining} > 0$, draw $X_t$ uniformly from the most negative states $\mathcal{S}_-$ with probability $p_{\text{neg}}$, otherwise from the most positive states $\mathcal{S}_+$. Set $\texttt{remaining} \leftarrow \texttt{remaining} - 1$.
3. Ordinary day: if $\texttt{remaining} = 0$, draw $X_t$ from row $X_{t-1}$ of $\hat{\mathbf{P}}$.
4. Emission: $g_t \leftarrow \mu_k + \sigma_k T_\nu$, where $X_t = s_k$.

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**Full version:** the lecture notebook

# Tuning and Checking the Model

A grid search picks $(\epsilon, \lambda)$. On the $M$ simulated paths with an episode, it scores each path's autocorrelation $\hat{\rho}_{m,k}$ of $|g_t|$ and kurtosis $\hat{\kappa}_m$ against the data's $\rho_k$, $\kappa$:

$$J(\epsilon, \lambda) = \frac{1}{M} \sum_{m=1}^{M} \Big[ \sum_{k=1}^{25} \left(\rho_k - \hat{\rho}_{m,k}\right)^2 + 0.2 \left(\kappa - \hat{\kappa}_m\right)^2 \Big]$$

The score is noisy, so the chosen pair can change with the seed.

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**Example:** CHEME-5660-L13a-Example-JumpHMM-SPY-Fall-2026.ipynb

# Review: Factor Models

Previously, we studied factor models, which explain how stocks move together with a few common factors:

$$g_i(t) = \underbrace{\alpha_i}_{\text{firm}} + \underbrace{\beta_i\, g_m(t)}_{\text{market}} + \underbrace{\sum_j \delta_{ij}\, f_j(t)}_{\text{other factors}} + \underbrace{\varepsilon_i(t)}_{\text{residual}}$$

The single index model keeps only the market factor. It assumes that the residuals of different stocks are uncorrelated, which means that only the market links two stocks. In the data, the residuals of stocks in the same sector, such as two banks, are correlated.

# Many Assets: The Sector Factor Model

**Definition:** in the sector factor model, each stock has one other factor besides the market, the factor $f_s$ of its sector $s$. The daily growth rate of stock $i$ in sector $s$ is given by:

$$g_i(t) = \alpha_i + \beta_i \underbrace{g_m(t)}_{\text{market}} + \delta_i \underbrace{f_s(t)}_{\text{sector}} + \psi_i \underbrace{\tilde{\varepsilon}_i(t)}_{\text{own}}$$

Here $\alpha_i$ and $\beta_i$ are the single index estimates, held fixed. The sector draw $f_s$ models the sector's average residual, with loading $\delta_i$. The own draw $\tilde{\varepsilon}_i$ models the stock's residual left after the sector term, centered and sized by $\psi_i$ so the sector and own terms keep the residual's variance. Each draw is a jump hidden Markov model, and the shared draws are the only link between stocks.

# What the Model Keeps and Gives Up

Composing stocks from shared draws on the same day has consequences:

- Keeps: fat tails and clustered volatility in every draw; stocks in a sector move together through their shared sector draw, all stocks through the market draw.
- Gives up: sector draws are independent, so stocks in different sectors are linked only through the market.
- No copula: a copula reorders each stock's draws across days and would scramble the timing of the volatile periods and jump episodes in them.

# An Option: A Drifting $\beta$

Holding $\beta_i$ fixed is a simplification: real $\beta$ wanders. The model can let $\beta_i(t)$ drift around $\hat{\beta}_i$ as an exponential moving average of random shocks, sized by how much the online estimate of $\beta$ wanders in the real data.

A trading method that re-estimates $\beta$ as it trades then faces the same problem in the synthetic market as in the real one.

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**Example:** CHEME-5660-L13a-Example-JumpHMM-SectorFactor-Fall-2026.ipynb

# Optional Advanced Material

**Markov models:** advanced/markov/
CHEME-5660-L13a-Lecture-MarkovModels-Fall-2026.ipynb

The Markov property, the stationary distribution by power iteration, and the sampling algorithm of a hidden Markov model. Two examples apply a hidden Markov model to a person's mood and to the daily growth rate of `SPY`.

# Summary

We built synthetic markets for stress tests, from Markov chains to a jump hidden Markov model of one asset and a sector factor model of many stocks.

## Key takeaways

- Markov chains: states, a transition matrix estimated by counting, and a stationary distribution; hidden states emit what we observe.
- Jump hidden Markov model: fat tails from binned Student-t states, and part of the volatility clustering from tuned jump episodes.
- Sector factor model: shared market and sector draws make synthetic stocks move together.

Synthetic markets let us test a method on many paths, not one.